

D.A.V. LOHAGHAT
CLASS TEST - II (2025-26)
SUBJECT - PHYSICS
CLASS-XII

MM : 30

Time: 1:00 hours

Section A - MCQ

- Q 1.** Two equal point charges each of $3 \mu\text{C}$ are separated by a certain distance in metres. If they are located at $(\hat{i} + \hat{j} + \hat{k})$ and $(2\hat{i} + 3\hat{j} + \hat{k})$ then the electrostatic force between them is [1]
- A. $9 \times 10^3 \text{ N}$ B. $16 \times 10^{-3} \text{ N}$ C. 10^{-3} N D. $9 \times 10^{-2} \text{ N}$
- Q 2.** In an experiment three microscopic latex spheres are sprayed into a chamber and became charged with charges $+3e$, $+5e$ and $-3e$ respectively. All the spheres came in contact simultaneously for a moment and got separated. Which one of the following possible values for the final charge on spheres? [1]
- A. $+5e, -4e, +5e$ C. $-4e, +3.5e, +5.5e$
B. $+6e, +6e, -7e$ D. $+5e, -8e, +7e$
- Q 3.** An electric dipole with dipole moment $4 \times 10^{-9} \text{ C-m}$ is aligned at 30° with the direction of a uniform electric field of magnitude $5 \times 10^4 \text{ NC}^{-1}$. The torque acting on the dipole is [1]
- A. $1 \times 10^{-4} \text{ Nm}$ B. $5 \times 10^{-8} \text{ Nm}$ C. $11 \times 10^{-12} \text{ Nm}$ D. $25 \times 10^{-19} \text{ Nm}$
- Q 4.** The electric field intensity just sufficient to balance the earth's gravitational attraction on an electron will be: (given mass and charge of an electron respectively are $9.1 \times 10^{-31} \text{ kg}$ and $1.6 \times 10^{-19} \text{ C}$.) [1]
- A. $-5.6 \times 10^{-11} \text{ N/C}$ C. $-1.6 \times 10^{-19} \text{ N/C}$
B. $-4.8 \times 10^{-15} \text{ N/C}$ D. $-3.2 \times 10^{-19} \text{ N/C}$
- Q 5.** A closed surface in vacuum encloses charges $-q$ and $+3q$. Another charge $-2q$ lies outside the surface. Total electric flux over the surface is [1]
- A. Zero B. $\frac{2q}{\epsilon_0}$ C. $\frac{-3q}{\epsilon_0}$ D. $\frac{4q}{\epsilon_0}$
- Q 6.** If the electric flux entering and leaving an enclosed surface respectively is ϕ_1 and ϕ_2 , the electric charge inside the surface will be [1]
- A. $(\phi_1 + \phi_2) \times \epsilon_0$ B. $(\phi_2 - \phi_1) \times \epsilon_0$ C. $(\phi_1 - \phi_2) \times \epsilon_0$ D. Zero

Q 7. Each of two point charges is doubles and their distance is halved. Force of interaction becomes n times, where n is [1]

A. 4

B. 1

C. 18

D. 16

Section B - Very Short Answer Questions

Q 8. Write any four properties of electric field lines. [2]

Q 9. A surface element $d\vec{S} = 20\hat{i}$ is placed in an electric field $\vec{E} = 4\hat{i} + 8\hat{j} + 14\hat{k}$. What is the electric flux emanating from the surface? [2]

Q 10. An infinite line charge produces a field of 9×10^4 N/C at a distance of 0.02 m. Calculate the linear charge density. [2]

Q 11. A charge of 17.7×10^{-4} C is distributed over a large sheet of area 400 cm^2 . Calculate the electric field intensity at a distance of 10 cm from it. [2]

Section C - Short Answer Questions

Q 12. Using Gauss theorem obtain an expression for electric field intensity at a point due to infinitely long line charge distribution. Sketch graphically variation of E with distance r. [3]

Q 13. Two point charges of $+1 \mu\text{C}$ and $+4 \mu\text{C}$ are kept 30 cm apart. How far from the $+1 \mu\text{C}$ charge on the line joining the two charges will the net electric field be zero? [3]

Q 14. State Gauss's theorem in electrostatics. How will you prove it for spherically symmetric surfaces? [3]

Q 15. (a) Obtain the expression for the torque experienced by an electric dipole of dipole moment $\vec{p}$ in a uniform electric field, $\vec{E}$. [3]

(b) What will happen, if the field were not uniform?

Q 16. Given the electric field in the region $\vec{E} = 2x\hat{i}$, find the net electric flux through the cube and the charge enclosed by it. [3]

